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Patrik Reizinger

Summary

AI is becoming an organic part of research, posing both a challenge and an opportunity: to redefine how we do research for the better.

To use AI in research, we need robust and trustworthy methods. To this end, my doctoral work builds identifiability theory for understanding *when and why neural networks generalize*, with syntheses of the overarching patterns across causality, self-supervised learning, language models, and interpretability. I believe that understanding should be turned into better tools, which I actively build, from benchmarks and open tooling for AI-assisted research to a public Challenge program for verifying AI-assisted science.

But tooling is not enough: the biggest leverage I can imagine is improving how we train exceptional researchers, for which I adopt best practices from coaching, leadership, and the psychology of high performance.

Education

2026.09–ongoing **Postdoctoral Fellow**, *Mila/Université de Montréal*, Montréal, Canada
Topic: Robust language models via identifiable causal mechanisms
Supervisors: Dhanya Sridhar, Simon Lacoste-Julien

2021–2026 **Machine Learning Ph.D.**, *International Max Planck Research School for Intelligent Systems/ELLIS Institute Tübingen*, Tübingen, Germany
Thesis: Causal Representation Learning
Supervisors: Wieland Brendel, Ferenc Huszár, Matthias Bethge, Bernhard Schölkopf
ELLIS exchange at the University of Cambridge: 2022.10–2023.03.

2019–2021 **Electrical Engineering M.Sc.**, *Budapest University of Technology and Economics*, Budapest, Hungary, GPA 5.0/5.0 (valedictorian)
Thesis: Development of an Attitude Determination and Control System for CubeSats on LEO orbits
Supervisors: Ferenc Vajda, Márton Szemenyei
Extracurricular: iMSc program for talented students

2015–2019 **Electrical Engineering B.Sc.**, *Budapest University of Technology and Economics*, Budapest, Hungary, GPA 5.0/5.0 (valedictorian)
Thesis: Development of a 3D input device for virtual working environments
Supervisors: Ferenc Vajda, Márton Szemenyei
Extracurricular: German language program in cooperation with the Karlsruhe Institute of Technology
Exchange semester at the Karlsruhe Institute of Technology: 2018.10–2019.02.

Publications

Selected

A unifying framework from neural superposition to sparse interpretable codes. D. Klindt, C. O'Neill, **Patrik Reizinger**, H. Maurer, N. Miolane. *Nature Machine Intelligence* 8(7):1025–1037 (2026).

Position: An Empirically Grounded Identifiability Theory Will Accelerate Self-Supervised Learning Research. **Patrik Reizinger**[*], R. Balestriero, D. Klindt, W. Brendel. *ICML 2025*.

Cross-Entropy Is All You Need To Invert the Data Generating Process. **Patrik Reizinger**[*], A. Bizeul[*], A. Juhos[*], J. E. Vogt, R. Balestriero, W. Brendel, D. Klindt. *ICLR 2025* — **Oral** (top 1.8%).

Identifiable Exchangeable Mechanisms for Causal Structure and Representation Learning. **Patrik Reizinger**[*], S. Guo[*], F. Huszár, B. Schölkopf, W. Brendel. *ICLR 2025* — **Spotlight**.

Rule Extrapolation in Language Modeling: A Study of Compositional Generalization on OOD Prompts. A. Mészáros, S. Ujváry, W. Brendel, **Patrik Reizinger**[†], F. Huszár[†]. *NeurIPS 2024* — **Spotlight**.

Position: Understanding LLMs Requires More Than Statistical Generalization. **Patrik Reizinger**[*], S. Ujváry[*], A. Mészáros[*], A. Kerekes[*], W. Brendel, F. Huszár. *ICML 2024* — **Spotlight** (top 3.5%).

Full list

[1] Karim Zaghw, Wieland Brendel[†], and **Patrik Reizinger**[†]. *When Do Learned Blocks Preserve Causal Meaning? Block Identifiability and Constructive Causal Abstractions*. 2026. arXiv: 2610.18155 [cs.LG].

[2] Kayoon Kim, Wieland Brendel[†], and **Patrik Reizinger**[†]. *HALLMARK: How to Diagnose LLM Citation Verifiers and When to Deploy Them*. 2026. arXiv: 2607.18360 [cs.CR].

[3] Hsun-Yu Kuo, El Mahdi Chayti, **Patrik Reizinger**, Wieland Brendel, and Martin Jaggi. "Stabilizing Extrapolation in Looped Transformers via Learned Stochastic Stopping". In: *The Fortieth Annual Conference on Neural Information Processing Systems*. 2026.

[4] David Klindt, Charles O'Neill, **Patrik Reizinger**, Harald Maurer, and Nina Miolane. "A unifying framework from neural superposition to sparse interpretable codes". In: *Nature Machine Intelligence* 8.7 (July 2026), pp. 1025–1037.

[5] Shruti Joshi, Aaron Mueller, David Klindt, Wieland Brendel, **Patrik Reizinger**[†], and Dhanya Sridhar[†]. "Position: Causality is Key for Interpretability Claims to Generalise". In: *Forty-third International Conference on Machine Learning Position Paper Track*. 2026.

[6] Anna Mészáros, **Patrik Reizinger**, and Ferenc Huszár. "Out-of-Distribution Evaluation of Rule-Based and Strategic Reasoning in Chess Transformers". In: *Forty-third International Conference on Machine Learning*. 2026.

[7] **Patrik Reizinger** and Wieland Brendel. "Skills, Benchmarks, and Verification Are What AI-Assisted Research Needs". In: *Post-AGI Science and Society Workshop*. 2026.

[8] Shruti Joshi, Théo Saulus, Wieland Brendel, Philippe Brouillard, Dhanya Sridhar, and **Patrik Reizinger**. "Who Guards the Guardians? The Challenges of Evaluating Identifiability of Learned Representations". In: *Forty-Second Annual Conference on Uncertainty in Artificial Intelligence*. 2026.

[9] Eric Gan*, **Patrik Reizinger***, Alice Bizeul*, Attila Juhos, Mark Ibrahim, Randall Balestriero, David Klindt, Wieland Brendel, and Baharan Mirzasoleiman. "Occam's Razor for SSL: Memory-Efficient Parametric Instance Discrimination". In: *Transactions on Machine Learning Research* (2025).

[10] Aaron Mueller, Andrew Lee, Shruti Joshi, Ekdeep Singh Lubana, Dhanya Sridhar, and **Patrik Reizinger**. "From Isolation to Entanglement: When Do Interpretability Methods Identify and Disentangle Known Concepts?" In: *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2026.

[11] **Patrik Reizinger**, Lester Mackey, Wieland Brendel, and Rahul Krishnan. *Estimating Treatment Effects with Independent Component Analysis*. 2025. arXiv: 2507.16467 [stat.ML].

[12] **Patrik Reizinger***, Bálint Mucsányi*, Siyuan Guo*, Benjamin Eysenbach, Bernhard Schölkopf†, and Wieland Brendel†. "Skill Learning via Policy Diversity Yields Identifiable Representations for Reinforcement Learning". In: *The Fourteenth International Conference on Learning Representations*. 2026. arXiv: 2507.14748 [cs.LG].

[13] **Patrik Reizinger**, Randall Balestriero, David Klindt, and Wieland Brendel. "Position: An Empirically Grounded Identifiability Theory Will Accelerate Self-Supervised Learning Research". In: *The 42nd International Conference on Machine Learning*. 2025.

[14] Szilvia Ujváry, Anna Mészáros, Wieland Brendel, **Patrik Reizinger**†, and Ferenc Huszár†. "Transcending Bayesian Inference: Transformers Extrapolate Rules Compositionally Under Model Misspecification". In: *7th Symposium on Advances in Approximate Bayesian Inference – Workshop Track*. 2025.

[15] **Patrik Reizinger***, Alice Bizeul*, Attila Juhos*, Julia E. Vogt, Randall Balestriero, Wieland Brendel, and David Klindt. "Cross-Entropy Is All You Need To Invert the Data Generating Process". In: *The Thirteenth International Conference on Learning Representations*. Oral (top 1.8%). 2025.

[16] **Patrik Reizinger***, Siyuan Guo*, Ferenc Huszár, Bernhard Schölkopf, and Wieland Brendel. "Identifiable Exchangeable Mechanisms for Causal Structure and Representation Learning". In: *The Thirteenth International Conference on Learning Representations*. Spotlight (top 5.1%). 2025.

[17] Evgenia Rusak*, **Patrik Reizinger***, Attila Juhos*, Oliver Bringmann, Roland S. Zimmermann, and Wieland Brendel. "InfoNCE: Identifying the Gap Between Theory and Practice". In: *The 28th International Conference on Artificial Intelligence and Statistics*. 2025.

[18] **Patrik Reizinger** and Rahul G. Krishnan. "Exploring A Bayesian View On Compositional and Counterfactual Generalization". In: *NeurIPS 2024 Workshop on Compositional Learning: Perspectives, Methods, and Paths Forward*. 2024.

[19] Anna Mészáros, Szilvia Ujváry, Wieland Brendel, **Patrik Reizinger**†, and Ferenc Huszár†. "Rule Extrapolation in Language Modeling: A Study of Compositional Generalization on OOD Prompts". In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. Spotlight. 2024.

[20] **Patrik Reizinger***, Szilvia Ujváry*, Anna Mészáros*, Anna Kerekes*, Wieland Brendel, and Ferenc Huszár. "Position: Understanding LLMs Requires More Than Statistical Generalization". In: *Forty-first International Conference on Machine Learning*. Spotlight (top 3.5%). 2024.

[21] Goutham Rajendran*, **Patrik Reizinger***, Wieland Brendel, and Pradeep Kumar Ravikumar. "An Interventional Perspective on Identifiability in Gaussian LTI Systems with Independent Component Analysis". In: *3rd Conference on Causal Learning and Reasoning (CLeaR)*. Oral. 2024.

[22] **Patrik Reizinger***, Han-Bo Li*, Aditya Ravuri*, Ferenc Huszár, and Neil D. Lawrence. "Independent Mechanism Analysis in GPLVMs". In: *Fifth Symposium on Advances in Approximate Bayesian Inference*. 2023.

[23] **Patrik Reizinger** and Ferenc Huszár. "SAMBA: Regularized Autoencoders perform Sharpness-Aware Minimization". In: *Fifth Symposium on Advances in Approximate Bayesian Inference*. 2023.

[24] Hamza Keurti*, **Patrik Reizinger***, Bernhard Schölkopf, and Wieland Brendel. "Desiderata for Representation Learning from Identifiability, Disentanglement, and Group-Structuredness". In: *ICML 2023 Workshop 2nd Annual TAG in Machine Learning*. 2023.

[25] **Patrik Reizinger** and Ferenc Vajda. "CubeSat Attitude Determination with Decomposed Kalman Filters". In: *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 45.2 (2023), p. 126.

[26] **Patrik Reizinger**. "The Falsificationist View of Machine Learning". In: *Information Society/Információs Társadalom (InfTars)* 23.2 (2023).

[27] **Patrik Reizinger**, Yash Sharma, Matthias Bethge, Bernhard Schölkopf, Ferenc Huszár, and Wieland Brendel. "Jacobian-based Causal Discovery with Nonlinear ICA". In: *Transactions on Machine Learning Research* (2023).

[28] Evgenia Rusak*, **Patrik Reizinger***, Roland S. Zimmermann*, Oliver Bringmann, and Wieland Brendel. "Content Suppresses Style in Dimensionality Collapse in Contrastive Learning". In: *NeurIPS 2022 Workshop on Self-Supervised Learning–Theory and Practice*. 2022.

[29] **Patrik Reizinger***, Luigi Gresele*, Jack Brady*, Julius von Kügelgen, Dominik Zietlow, Bernhard Schölkopf, Georg Martius, Wieland Brendel, and Michel Besserve. "Embrace the Gap: VAEs Perform Independent Mechanism Analysis". In: *NeurIPS 2022*. 2022.

[30] **Patrik Reizinger***, Yash Sharma, Matthias Bethge, Bernhard Schölkopf, Ferenc Huszár, and Wieland Brendel. "Multivariable Causal Discovery with General Nonlinear Relationships". In: *UAI 2022 Workshop on Causal Representation Learning*. Oral. 2022.

[31] Márton Szemenyei and **Patrik Reizinger**. "Handling Realistic Noise in Multi-Agent Systems with Self-Supervised Learning and Curiosity". In: *Journal of Artificial Intelligence and Soft Computing Research* 12.2 (2021), pp. 135–148.

[32] **Patrik Reizinger**, Péter Huszár, Dorottya Milánkovich, and Alexandra Széll. "Kisműholdak fejlesztése a sokoldalúság és a könnyű reprodukálhatóság tükrében". In: *Repüléstudományi Közlemények* 32.2 (2020), pp. 81–95.

[33] Márton Szemenyei and **Patrik Reizinger**. "Learning to Play Robot Soccer from Partial Observations". In: *2020 23rd International Symposium on Measurement and Control in Robotics (ISMCR)*. IEEE. 2020, pp. 1–6.

[34] **Patrik Reizinger** and Márton Szemenyei. "Attention-based curiosity-driven exploration in deep reinforcement learning". In: *ICASSP 2020 – 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2020, pp. 3542–3546.

[35] Marton Szemenyei and **Patrik Reizinger**. "Attention-based curiosity in multi-agent reinforcement learning environments". In: *2019 International Conference on Control, Artificial Intelligence, Robotics & Optimization (ICCAIRO)*. IEEE. 2019, pp. 176–181.

[36] **Patrik Reizinger** and Bálint Gyires-Tóth. "Stochastic weight matrix-based regularization methods for deep neural networks". In: *International Conference on Machine Learning, Optimization, and Data Science*. Springer, Cham. 2019, pp. 45–57.

[37] **Patrik Reizinger** and Ferenc Vajda. *Concept of a mobile, cameraless VR-controller framework for working environments*. 2018.

* denotes joint first, † joint senior authorship.

Experience

2025.10–12. **Visiting Researcher**, *Mila*, Montréal, Canada
Hosted by Dhanya Sridhar and Simon Lacoste-Julien, working on causal representation learning, mechanistic interpretability, LLMs, and optimization

2024.06–09. **Summer Research Intern**, *Vector Institute*, Toronto, Canada
Hosted by Rahul G. Krishnan, working on causal representation learning and self-supervised methods

2020.02–2021.01. **Deep Learning Student Researcher**, *Budapest University of Technology and Economics*, Budapest, Hungary
Analyzed time series data with deep learning

2019.02–2021.02. **Control Engineering Intern**, *C3S Electronics Development LLC*, Budapest, Hungary
Developed and designed a CubeSat attitude determination and control system

2019.01–02. **FPGA Developer Intern**, *Karlsruhe Institute of Technology*, Karlsruhe, Germany
Implemented FPGA time synchronisation with a Python interface

2018.06–08. **Image Processing Intern**, *Fraunhofer Institute for Factory Operation and Automation IFF*, Magdeburg, Germany
Developed an automated visual inspection tool in C++, including a Python wrapper

2017.06–08. **Data Scientist Intern**, *Gravity R&D*, Budapest, Hungary
Analyzed customer data in Python with machine learning

2016.09–2019.01. **Virtual Reality Peripheral Device Developer**, *Budapest University of Technology and Economics*, Budapest, Hungary
Developed the hardware and software for a 3D input device for virtual working environments

Honors and Awards

2026 **Fellow, Big-if-True Science Accelerator**, *Renaissance Philanthropy & SPRIND*, Second cohort, 8 fellows, Designing a public Challenge program for AI-assisted verification of science, mentored by DARPA and ARPA-H program managers

2023, 2024 **Qualcomm Innovation Fellowship Europe**, *Qualcomm*, Finalist, AI/cybersecurity fellowship; finalists present to Qualcomm judges

2021 **Pro Scientia Gold Medal**, *National Scientific Students' Association Hungary*, top 0.03%, Highest national student research distinction for overall scientific excellence

2021 **Hope Badge Special Award to The Most Promising Young Scientist**, *Pro Scientia Gold Medalists' Association*, top 0.003%, Highest national student research distinction at the National Scientific Students' Association Conference

2019, 2021 **1st Prize at National Scientific Students' Association Conference**, *National Scientific Students' Association Hungary*, top 0.3%

- Attention-based curiosity in multi-agent reinforcement learning environments (2021)
- Stochastic weight matrix-based regularization methods for deep neural networks (2019)
- Development of a 3D input device for virtual working environments (2019)

2017–2019 **1st Prize at Scientific Students' Association Conference**, *Budapest University of Technology and Economics*, top 0.3%

- Attention-based curiosity in multi-agent reinforcement learning environments (2019)
- Development of an Attitude Determination and Control System for CubeSats on LEO orbits (2019)
- Stochastic weight matrix-based regularization methods for deep neural networks (2018)
- Development of a 3D input device for virtual working environments (2017)

2018 **Nokia Bell Labs Scholarship for Deep Learning Research**, *Nokia Bell Labs Hungary*, Early research scholarship from a globally recognized industrial lab

2016–2018, 2020 **National Higher Education Scholarship**, *Republic of Hungary*, top 0.8%, Highest national student distinction for academic excellence

Grants and Funding

2026 **IVADO Postdoctoral Research Fellowship**, *Canada First Excellence Research Fund*, CAD 100,000/year for 2 years

2026 **Canada Impact+ Postdoctoral Research Training Award**, *Natural Sciences and Engineering Research Council of Canada*, CAD 140,000 for 2 years, Declined

2025 **ICLR Scholar Award**, *Conference support for accepted authors*

2022–2023 **ELLIS Travel Grant**, *Travel support from the ELLIS network for my exchange at the University of Cambridge*

2022 **NeurIPS Scholar Award**, *Conference support for accepted authors/participants*

2016, 2018 **New National Excellence Program Research Grant**, *Ministry of Innovation and Technology Hungary*, top 0.3%, Competitive research grant for university students

Talks

2026.10. **Causal Foundations for Interpretability — What Your Methods Actually Identify**, *Nagymaros AI Retreat*, Nagymaros, Hungary

2026.10. **Synergies between Mechanistic Interpretability and Causal Representation Learning**, *BIRS Workshop on Identifiable Representation Learning*, Banff, Canada, Invited

2026.05. **Developing Research Taste in the Era of AI-assisted Research**, *BethgeLab Compounding Intelligence Workshop*, Tübingen, Germany

2025.11. **Rule Extrapolation in Language Models**, *Vector Institute*, Toronto, Canada, Invited

2025.10. **Identifiable Exchangeable Mechanisms**, *Symposium on Mathematical Foundations of Trustworthy Learning*, Ascona, Switzerland

2025.09. **Identifiable Exchangeable Mechanisms**, *Centre Suisse d'Electronique et de Microtechnique*, Alpnach, Switzerland, Invited

2025.09. **Causal Representation Learning**, *Laboratory of Cryptography and System Security*, Budapest, Hungary, Invited

2025.08. **Identifiable Exchangeable Mechanisms**, *Hungarian Machine Learning Days, Artificial Intelligence National Laboratory*, Budapest, Hungary, Invited

2025.08. **Estimating Treatment Effects with Independent Component Analysis**, *4th MCML Workshop on Causal Machine Learning, MCML*, München, Germany, Invited

2025.02. **Causality and OOD generalization in Foundation Models**, *Bellairs Workshop on Causality, Bellairs Research Institute*, Barbados, Invited

2024.09. **Identifiable Exchangeable Mechanisms**, *Mila tea talk series, Mila*, Montréal, Canada, Invited

2024.07. **Embrace the Gap: VAEs Perform Independent Mechanism Analysis**, *Critical ML Lab, University of Waterloo*, Waterloo, Canada, Invited

2023.12. **Embrace the Gap: VAEs Perform Independent Mechanism Analysis**, *Central European University Representation Learning Reading Group*, Budapest, Hungary, Invited

2023.02. **Popper meets machine learning—How falsificationism can guide the design of AI solutions**, *Darwin College*, Cambridge, UK

2023.01. **Multivariable Causal Discovery for General Nonlinear Functions**, *AstraZeneca Seminar*, online, Invited

2022.12. **Multivariable Causal Discovery for General Nonlinear Functions**, *Learning on Graphs Cambridge Meetup*, Cambridge, UK

2022.10. **Embrace the Gap: VAEs Perform Independent Mechanism Analysis**, *University of Warsaw Machine Learning Seminar*, online, Invited

2022.08. **Multivariable Causal Discovery for General Nonlinear Functions**, *UAI 2022 Workshop on Causal Representation Learning*, Eindhoven, Netherlands

Service & Outreach

2026 **Organizer**, *Interpretability as a Science: Toward Rigorous Foundations for Understanding LLMs*, NeurIPS 2026

2026 **Organizer**, *Tutorial: Causal Foundations for Interpretability — What Your Methods Actually Identify*, EMNLP 2026
Selected from a record 56 proposals (25% accepted across EMNLP and AACL)

2024 **Organizer**, *CALM: First Workshop on Causality and Large Models*, NeurIPS 2024

2023, 2024 **Organizer**, *IMPRS-IS Bootcamp*, International Max Planck Research School for Intelligent Systems

2022, 2023 **Organizer**, *ELLIS Doctoral Symposium*, ELLIS

2022–ongoing **Academic Knowledge Transfer**, *The Path to PhD blog*, path2phd.substack.com

2021–ongoing **Thesis committee member**, *Budapest University of Technology and Economics*

2020–2025 **Coordinator**, *Machine Learning Journal Club for Hungarian Students*

2020 **Program Committee Member**, *6th International Conference on Machine Learning, Optimization, and Data Science*

2016 **E-learning Developer**, *EduBase*
Video series for the Digital Design I. university course

Reviewing JMLR; ICML 2025/2026 and NeurIPS 2026 position paper tracks; ICLR 2025/2026/2027; NeurIPS 2024/2025; CLeaR 2024/2025; AISTATS 2026/2027; UAI 2026; NeurIPS 2023/2024 workshops; Infocommunications Journal

Open-source tools HALLMARK — benchmark for citation-hallucination detection in scientific writing
`bibtexupdater` — automatic preprint-to-published BibTeX migration and reference validation
Research Agora — an open marketplace for AI-assisted research skills

MENTORING

2026–ongoing **Karim Zaghw**, *M.Sc. at the University of Tübingen*

2026–ongoing **Kayoon Kim**, *M.Sc. at the University of Tübingen*

2025–ongoing **Hsun-Yu Kuo**, *M.Sc. at EPFL*

2025 **Samuel Innes**, *B.Sc. at Heidelberg → M.Sc. at the University of Cambridge*

2023–ongoing **Bálint Mucsányi**, *M.Sc. → Ph.D. at the University of Tübingen*

2023–ongoing **Szilvia Ujváry**, *M.Sc. → Ph.D. at the University of Cambridge*

2023–ongoing **Anna Mészáros**, *M.Sc. → Ph.D. at the University of Cambridge*

2022–2025 **Boglárka Ecsedi**, *B.Sc. at GeorgiaTech → Ph.D. at the University of Toronto*

TEACHING

2026.10. **Tutorial: Causal Foundations for Interpretability — What Your Methods Actually Identify**, *EMNLP 2026*, Budapest, Hungary
Half-day tutorial with Shruti Joshi, Aaron Mueller, and Dhanya Sridhar

2026.09. **Tutorial: Principles for (Agentic) Research**, *50th Machine Learning Summer School (MLSS)*, Tübingen, Germany
2h tutorial

2026.03. **Tutorial: AI Agent Tool Use for PIs**, *ML Excellence Cluster Tübingen Retreat*, Heiligkreuztal, Germany
Hands-on 2h workshop for ∼ 30 PIs across ML, neuroscience, and the humanities

Fall 2020/21 **Teaching Assistant**, *Budapest University of Technology and Economics*, Budapest, Hungary
Image Processing Laboratory I., Computer Vision Systems, Deep Learning in Visual Computing

Fall 2017/18 **Teaching Assistant**, *Budapest University of Technology and Economics*, Budapest, Hungary
Digital Design I. laboratory

RESEARCH & TECHNICAL TRAINING

2026.08. **International Summer School Marktoberdorf — Engineering Secure and Dependable Software Systems**, *TU München & fortiss*, Herrsching, Germany

2023.12. **CI/CD for Machine Learning certification**, *Weights and Biases*, online

2022.07. **ELLIS Cambridge Unit Machine Learning Summer School**, *ELLIS Cambridge Unit*, Cambridge, UK

2022.07. **Machine Learning Summer School**, *ML in PL*, Krakow, Poland

2020.09. **Ladybird Guide to Spacecraft Operations Workshop**, *European Space Agency*, online

2020.07. **Eastern European Machine Learning Summer School**, *ML in PL*, online

2019.07. **International Summer School on Deep Learning**, *IRDTA*, Warsaw, Poland

2019.01. **Concurrent Design Workshop**, *European Space Agency*, ESEC-Galaxia, Redu, Belgium

Leadership & Personal Development

2024.07–ongoing **Performance Psychology, Leadership**, *The Growth Equation Academy*, online

2023.11–ongoing **Coaching, Exercise Physiology, Biomechanics**, *The Scholar Program*, online

2021.04. **A Young Leader's Guide to Risk**, *McChrystal Group*, Budapest, Hungary

2018–2020 **Leadership Academy**, *Mathias Corvinus Collegium*, Budapest, Hungary

2018.11. **Traction Europe Case Studies for Outstanding Engineering Students**, *Boston Consulting Group*, Paris, France

2016–2018 **Business and Economics Specialization**, *Mathias Corvinus Collegium*, Budapest, Hungary

2015–2016 **University Junior Program**, *Mathias Corvinus Collegium*, Budapest, Hungary

Competencies

Soft skills **Leadership** (co-organizing NeurIPS workshops in 2024 and 2026; IMPRS-IS Bootcamp; ELLIS Doctoral Symposium); **Program design** (Big-if-True Science Accelerator fellow, designing a public Challenge program under DARPA and ARPA-H program-manager mentorship); **Mentoring** (eight students since 2022, four of whom continued to a Ph.D.); **Scientific writing** (*Nature Machine Intelligence*; *Path to PhD* newsletter; reviewing for JMLR/ICML/ICLR/NeurIPS/CLeaR/AISTATS/UAI); **Project management** (leading multi-institution research collaborations across Mila, Vector Institute, IMPRS-IS, and Cambridge); **Assertive communication** (three ICML position papers staking out research directions); **Teaching & storytelling** (EMNLP 2026 and MLSS 2026 tutorials; 12 invited talks since 2022 at Mila, Vector Institute, Waterloo, et al.)

Languages English (C2), German (C2), Hungarian (native), Italian (A2), French (A1)

Montréal, 2nd November, 2026